SIAM EV Platform

Energy Management System (EMS) API Reference
External Integration Edition

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1 Introduction

This document describes the REST and WebSocket APIs for the **Energy Management System (EMS)** sub-system of the SIAM EV Unified Energy & Logistics Management Platform. It is a scoped extract of the full platform API reference, prepared for handover to a development partner responsible for the EMS facility/energy management system. It does not include the OCPP Charging Monitoring or Cargo Space Monitoring APIs, which remain internal to SIAM EV; see Section 6 for what a partner integrator needs to know about those boundaries.

The EMS sub-system aggregates energy consumption data from buildings and facilities, provides time-series analytics, and exposes demand-response controls. Data is persisted in TimescaleDB and read from SCADA / BMS systems over Modbus TCP or MQTT.

A *facility* is a physical site (office, retail, warehouse, or logistics hub). A *zone* is a named, bounded space inside a facility (cold room, loading dock, dry-storage bay) with its own energy draw and environmental sensors. EMS is the authoritative owner of all zone-level environmental and energy sensor data on the platform.

1.1 Base URL

All endpoints are served under the API Gateway. Replace `{host}` with the deployment address (e.g. `api.siam-ev.example.com`).

```
https://{host}/api/v1
```

Listing 1: Base URL pattern

EMS base path: `/api/v1/ems`

1.2 Authentication

Every request must carry a Bearer token obtained from the Auth Service.

```
Authorization: Bearer <access_token>
```

Listing 2: Authorization header

Header	Required	Description
`Authorization`	Yes	`Bearer <JWT>` issued by `/auth/token`
`Content-Type`	Yes (POST/PUT)	`application/json`
`X-Request-ID`	No	Idempotency key; echoed in response headers

1.3 Common Error Schema

```json
{
  "success": false,
  "error": {
    "code": "FACILITY_NOT_FOUND",
    "message": "No facility with id 'site-bkk-099' exists.",
    "timestamp": "2026-07-02T10:15:30Z"
  }
}
```

Listing 3: Error response envelope

HTTP Status	Meaning	Common cause
200	OK	Successful read
201	Created	Resource created
400	Bad Request	Missing or invalid fields
401	Unauthorized	Missing / expired token
403	Forbidden	Insufficient role
404	Not Found	Resource does not exist
409	Conflict	Duplicate resource
500	Internal Error	Upstream / database failure

2 Data Models

2.1 Object Model Overview

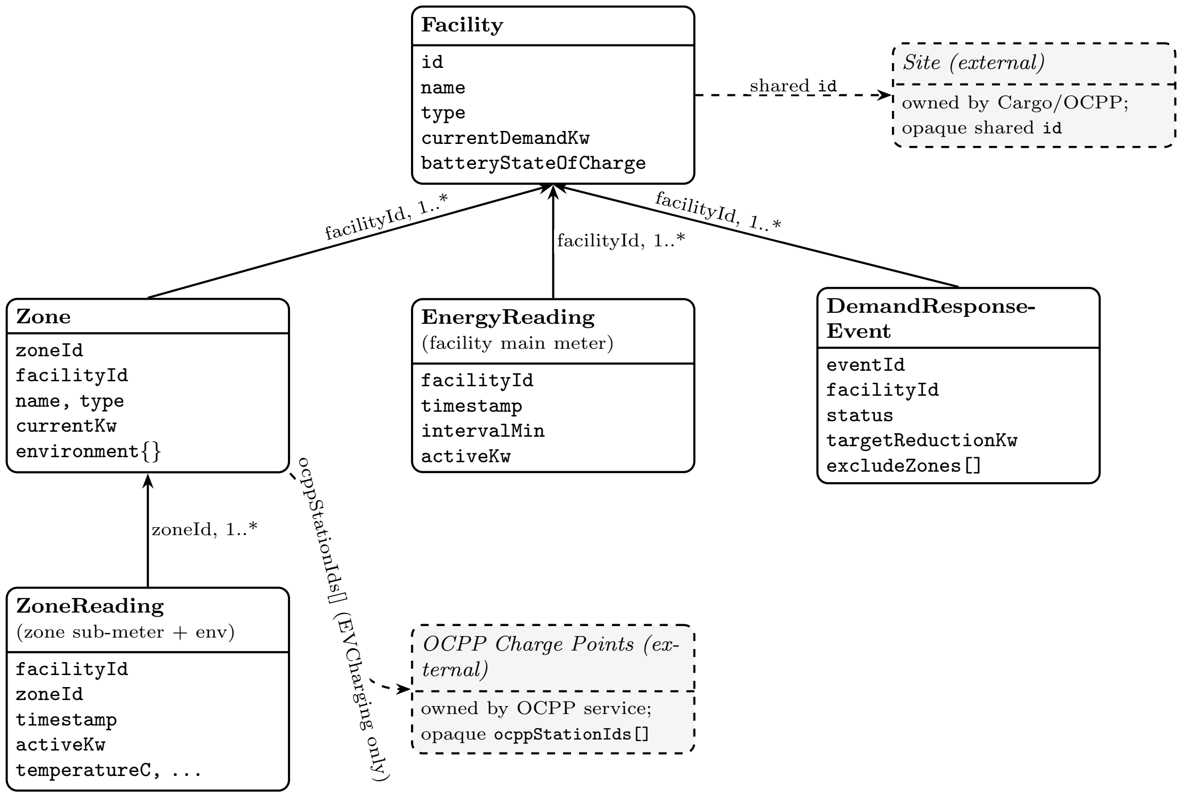

Figure 1: EMS object model — solid arrows are foreign-key references within this API. Note the two separate meter streams: `EnergyReading` is the facility's main meter (keyed by `facilityId` only), while `ZoneReading` is each zone's own sub-meter plus environment sensors (keyed by `facilityId` *and* `zoneId`) — one does not derive from the other. `DemandResponseEvent.excludeZones[]` also references Zone by ID. A `Zone` of `type: EVCharging` additionally references a group of OCPP charge points via `ocppStationIds[]`. Dashed boxes mark data whose owning system is out of scope for this document (see Section 6).

2.2 Facility Object

```
{
  "id":           "site-bkk-001",
  "name":         "Thai Post Sorting Hub --- Lad Krabang",
  "type":         "LogisticsWarehouse",  // Office | Retail | Warehouse | LogisticsWarehouse
  "location": {
    "lat": 13.7284,
    "lng": 100.7498,
    "address": "999 Moo 1, Lad Krabang, Bangkok 10520"
  },
  "ratedCapacityKva":    2000,
  "currentDemandKw":     312.4,
  "peakDemandKw":        1540.0,        // rolling 12-month peak
  "dailyEnergyKwh":      7820.5,
  "solarCapacityKw":     500,
  "batteryCapacityKwh":  1000,
  "batteryStateOfCharge": 0.78,         // 0.0--1.0
  "gridImportKw":        -45.2,         // negative = exporting to grid
  "demandResponseActive": false,
  "tariffZone":          "TOU-EV",
  "updatedAt": "2026-07-02T10:00:00Z"
}
```

Listing 4: Facility object schema

Note: some facility IDs correspond to physical sites that are also tracked by other internal SIAM EV systems (e.g. a logistics hub with its own fleet operations). That correspondence is maintained internally and does not appear in this schema; see Section 6.

2.3 Zone Object (authoritative sensor record)

A *zone* is a named, bounded space inside a facility (cold room, loading dock, dry-storage bay, EV charging area). All environmental sensor data for zones is owned by EMS and read from the facility's BMS or SCADA system over Modbus TCP or MQTT.

```
{
  "zoneId":      "zone-site-001-cold-01",
  "facilityId":  "site-bkk-001",
  "name":        "Cold Room A",
  "type":        "ColdRoom",    // ColdRoom | LoadingDock | DryBay | StagingArea | EVCharging
  // --- Energy (from SCADA / smart sub-meter) ---
  "currentKw":   42.1,
  "dailyKwh":    892.6,
  // --- Environment (from BMS sensors) ---
  "environment": {
    "temperatureC": 3.8,               // null if no temp sensor
    "humidityPct":  84.1,
    "co2Ppm":       405,
    "lightLux":     0
  },
  "setpointC":   4.0,                  // BMS setpoint (ColdRoom only)
  "thresholds": {
    "tempMinC":       2.0,
    "tempMaxC":       8.0,
    "humidityMaxPct": 90.0,
    "co2MaxPpm":      1000
  },
  "ocppStationIds": null,              // EVCharging only; see note below
  "alertActive": false,
  "source":      "BMS",                // BMS | SCADA | SmartMeter | Estimated
  "updatedAt":   "2026-07-02T10:02:00Z"
}
```

Listing 5: EMS zone object schema

```
{
  "zoneId":       "zone-site-001-ev-01",
  "facilityId":   "site-bkk-001",
  "name":         "EV Charging Bay 1",
  "type":         "EVCharging",
  "currentKw":    96.4,                  // sum of the linked chargers' live draw
  "dailyKwh":     1204.7,
  "environment": { "temperatureC": null, "humidityPct": null, "co2Ppm": null, "lightLux": null },
  "setpointC":    null,
  "thresholds":   null,                  // EVCharging zones use OCPP-side limits, not BMS thresholds
  "ocppStationIds": ["station-bkk-001", "station-bkk-002"], // the EVSE group metered by this zone
  "alertActive": false,
  "source":       "SmartMeter",
  "updatedAt":    "2026-07-02T10:02:00Z"
}
```

Listing 6: EMS zone object schema — EVCharging zone example

Note on `ocppStationIds`: an `EVCharging` zone groups a set of EVSEs/charge points for the purposes of sub-metering and demand-response shedding; `currentKw` is the aggregate live draw of that group. The station IDs themselves are opaque identifiers issued and owned by the OCPP charging system, which is outside the scope of this document — treat them as strings for display/filtering and do not expect to resolve charger-level detail (connector status, session state, etc.) from the EMS API. See Section 6.

2.4 Energy Reading Object (Time-Series) — Facility Main Meter

Keyed by `facilityId` only. This is the whole-site reading from the facility's main incoming meter, returned by `GET /ems/facilities/{id}/readings`.

```
{
  "facilityId":    "site-bkk-001",
  "timestamp":     "2026-07-02T10:00:00Z",
  "intervalMin":   15,                     // granularity of reading
  "activeKw":      315.2,
  "reactiveKvar": 42.1,
  "apparentKva":   317.9,
  "powerFactor":   0.991,
  "phaseA_V":      228.4,
  "phaseB_V":      229.1,
  "phaseC_V":      228.8,
  "source":        "SCADA"                // SCADA | BMS | SmartMeter | Estimated
}
```

Listing 7: Facility-level (main meter) energy reading schema

2.5 Zone Reading Object (Time-Series) — Zone Sub-Meter

Keyed by `facilityId` *and* `zoneId`. This is a distinct reading stream from the facility main meter above: each zone with its own sub-meter or BMS sensor reports its own energy draw, combined with that zone's environmental sensors in the same record. Returned by `GET /ems/facilities/{id}/zones/{zoneId}/readings`.

```
{
  "facilityId":    "site-bkk-001",
  "zoneId":        "zone-site-001-cold-01",
  "timestamp":     "2026-07-02T09:00:00Z",
  "intervalMin":   15,
  "activeKw":      44.8,                   // this zone's sub-meter, not the facility total
  "temperatureC": 3.9,
  "humidityPct":   85.2,
  "co2Ppm":        412
}
```

Listing 8: Zone-level (sub-meter + environment) reading schema

Note: a facility's main-meter `activeKw` is not simply the sum of its zones' sub-meter readings — common-area loads (lighting, HVAC plant, office space) are metered only at the facility level and have no corresponding zone.

3 Facility Endpoints

3.1 GET /ems/facilities — List facilities

Query parameters:

Param	Type	Default	Description
type	string	—	Filter by facility type
limit	integer	50	Max records
offset	integer	0	Pagination offset

Response 200:

```
{
  "success": true,
  "data": [ /* array of Facility objects */ ],
  "meta": { "total": 8, "limit": 50, "offset": 0 }
}
```

3.2 GET /ems/facilities/{id} — Get facility details

Returns the latest live snapshot for a single facility.

Response 200:

```
{ "success": true, "data": { /* Facility object */ } }
```

3.3 GET /ems/facilities/{id}/readings — Query time-series data

Returns historical energy readings for a facility over a given time range.

Query parameters:

Param	Type	Required	Description
from	ISO-8601	Yes	Start of range (UTC)
to	ISO-8601	Yes	End of range (UTC)
interval	string	No	Bucket size: 15m | 1h | 1d (default 15m)
metric	string	No	Specific field to return (e.g. activeKw)

Response 200:

```
{
  "success": true,
  "data": {
    "facilityId": "fac-bkk-001",
    "interval":   "1h",
    "readings": [
      { "timestamp": "2026-07-02T00:00:00Z", "activeKw": 180.4, "powerFactor": 0.98 },
      { "timestamp": "2026-07-02T01:00:00Z", "activeKw": 162.1, "powerFactor": 0.97 }
    ]
  }
}
```

Listing 9: Time-series readings response

3.4 GET /ems/facilities/{id}/analytics — Demand analytics

Returns aggregated KPIs for the facility over a billing period.

Query parameters: `period` — `day` — `week` — `month` — `year` (default `month`)

Response 200:

```
{
  "success": true,
  "data": {
    "facilityId":          "fac-bkk-001",
    "period":              "month",
    "totalConsumptionKwh": 234156.2,
    "peakDemandKw":         1540.0,
    "peakTimestamp":        "2026-06-14T14:30:00Z",
    "avgLoadFactorPct":     42.8,
    "solarGenerationKwh":   31200.0,
    "selfConsumptionPct":   88.4,
    "gridImportKwh":        202956.2,
    "gridExportKwh":        3600.0,
    "estimatedCostThb":     812400.50,
    "carbonKgCo2e":         91188.0
  }
}
```

Listing 10: Analytics response

3.5 PUT /ems/facilities/{id}/power-limit — Set import limit

Caps the maximum grid import power draw for a facility.

Request body:

```
{
  "limitKw":     800,
  "reason":      "PeakDemandAvoidance",  // PeakDemandAvoidance | GridRequest | Manual
  "expiresAt":   "2026-07-02T16:00:00Z" // null = permanent until changed
}
```

Response 200:

```
{ "success": true, "data": { "facilityId": "fac-bkk-001", "limitKw": 800, "appliedAt": "..."
    } }
```

3.6 POST /ems/facilities/{id}/demand-response — Trigger demand response

Activates or deactivates a demand-response event for a facility (e.g. shed non-critical loads to avoid peak demand charges).

Request body:

```
{
  "action":          "Activate",          // Activate | Deactivate
  "targetReductionKw": 200,               // required for Activate
  "durationMinutes":   30,                // required for Activate
  "priority":        "High",              // High | Medium | Low
  "notifyOccupants": true
}
```

Listing 11: Demand-response command

Response 200:

```
{
  "success": true,
  "data": {
    "eventId":           "dr-20260702-001",
    "status":            "Activated",
    "activatedAt":       "2026-07-02T14:00:00Z",
```

```
    "expiresAt":        "2026-07-02T14:30:00Z",
    "projectedSavingKw": 195.3
  }
}
```


Note: the request body may optionally include an `excludeZones` array of zone IDs to protect from shedding. Zone protection decisions are made by SIAM EV's internal optimizer; the EMS system only needs to honor whatever `excludeZones` list it receives.

4 Zone Endpoints

4.1 GET /ems/facilities/{id}/zones — List zones with sensor data

Returns all zones within a facility with their current environmental readings (temperature, humidity, CO_2) and energy draw. This is the **authoritative source** for zone sensor data on the platform.

Response 200:

```json
{
  "success": true,
  "data": {
    "facilityId":     "site-bkk-001",
    "totalCurrentKw": 312.4,
    "zones": [
      {
        "zoneId":      "zone-site-001-cold-01",
        "name":        "Cold Room A",
        "type":        "ColdRoom",
        "currentKw":   42.1,
        "dailyKwh":    892.6,
        "environment": { "temperatureC": 3.8, "humidityPct": 84.1, "co2Ppm": 405 },
        "setpointC":   4.0,
        "alertActive": false
      },
      {
        "zoneId":    "zone-site-001-dock-03",
        "name":      "Loading Dock 3",
        "type":      "LoadingDock",
        "currentKw": 12.7,
        "dailyKwh":  84.3,
        "environment": { "temperatureC": 31.2, "humidityPct": 62.0, "co2Ppm": 510 },
        "alertActive": false
      },
      {
        "zoneId":    "zone-site-001-ev-01",
        "name":      "EV Charging Bay 1",
        "type":      "EVCharging",
        "currentKw": 96.4,
        "dailyKwh":  1204.7,
        "environment": { "temperatureC": null, "humidityPct": null, "co2Ppm": null },
        "ocppStationIds": ["station-bkk-001", "station-bkk-002"],
        "alertActive": false
      }
    ]
  }
}
```

Listing 12: Zone list with sensor and energy data

Note: for some zone types, other internal SIAM EV systems may associate additional operational context with a zone (e.g. dock occupancy, or connector-level charging session detail for an `EVCharging` zone). That context is not part of the EMS contract and is joined internally by downstream consumers — it does not need to be produced by this endpoint.

4.2 GET /ems/facilities/{id}/zones/{zoneId} — Single zone detail

Returns the full EMS Zone object for one zone, including the complete threshold configuration and latest sensor readings.

Response 200:

```json
{ "success": true, "data": { /* EMS Zone object */ } }
```

4.3 GET /ems/facilities/{id}/zones/{zoneId}/readings — Zone sensor and energy history

Returns time-series readings for a single zone, combining energy (kW) and environmental sensors (temperature, humidity, CO_2) in one response. Accepts the same `from/to/interval` parameters as the facility-level readings endpoint.

Response 200:

```json
{
  "success": true,
  "data": {
    "facilityId": "site-bkk-001",
    "zoneId":     "zone-site-001-cold-01",
    "interval":   "1h",
    "readings": [
      { "timestamp": "2026-07-02T08:00:00Z",
        "activeKw": 40.2, "temperatureC": 4.1, "humidityPct": 83.5, "co2Ppm": 398 },
      { "timestamp": "2026-07-02T09:00:00Z",
        "activeKw": 44.8, "temperatureC": 3.9, "humidityPct": 85.2, "co2Ppm": 412 }
    ]
  }
}
```

Listing 13: Zone sensor + energy time-series

4.4 PUT /ems/facilities/{id}/zones/{zoneId}/thresholds — Update zone thresholds

Sets the alert thresholds for a zone. Alerts from this endpoint surface on the EMS WebSocket (`AlertTriggered`) and may also be surfaced to other internal SIAM EV dashboards; that distribution is handled internally and requires no action from the EMS system beyond emitting the WebSocket event.

Request body:

```json
{
  "tempMinC":       2.0,
  "tempMaxC":       8.0,
  "humidityMaxPct": 90.0,
  "co2MaxPpm":      1000
}
```

Response 200:

```json
{ "success": true, "data": { "zoneId": "zone-site-001-cold-01", "thresholds": { ... } } }
```

5 WebSocket — Live Energy Feed

```
wss://{host}/api/v1/ems/ws?facilityId=fac-bkk-001&token=<access_token>
```

Server-sent event frame:

```
{
  "event":      "EnergyReading",
  "facilityId": "fac-bkk-001",
  "data": {
    "activeKw":     318.7,
    "solarKw":       52.1,
    "batteryKw":    -10.0,    // negative = discharging (exporting to facility)
    "gridKw":       276.6,
    "soc":           0.74,
    "timestamp":    "2026-07-02T10:05:00Z"
  }
}
```

Listing 14: EMS live push event

event	Trigger
`EnergyReading`	Every 15 s from SCADA/BMS poller
`DemandAlert`	Demand approaches >90% of contract capacity
`SolarForecast`	Hourly PV generation forecast update
`BatteryStateChange`	SoC crosses 20%, 50%, or 80% threshold
`DemandResponseEvent`	DR event activated or deactivated
`AlertTriggered`	Zone threshold breach (see zone thresholds endpoint)
`AlertResolved`	Zone reading returns within threshold

6 Integration Notes

This section summarizes, at a high level, how EMS fits into the wider SIAM EV platform. Details of the other sub-systems are intentionally excluded from this handover document.

- **Facility identity is shared internally.** A `facilityId` (e.g. `site-bkk-001`) may correspond to the same physical site in other internal SIAM EV systems (e.g. a logistics hub tracked by a separate fleet-management system). This cross-reference is coordinated entirely on the SIAM EV side; the EMS API contract in this document does not change based on it, and no additional integration work is required by the EMS system to support it.
- **Zone protection during demand response.** SIAM EV's internal optimizer may ask EMS to exclude specific zones from a shed event via the `excludeZones` field on `POST /ems/facilities/{id}/demand-response`. The EMS system does not need to know why a zone is protected — only to honor the list it receives.
- **Smart charging coordination.** SIAM EV may adjust EV charger power limits in response to EMS demand alerts. This coordination is performed by an internal optimizer that consumes the EMS WebSocket feed and calls `PUT /ems/facilities/{id}/power-limit` to register the applied cap; it requires no changes to the EMS API.
- **`EVCharging` zones reference a group of OCPP charge points.** A zone of `type: "EVCharging"` carries an `ocppStationIds[]` array (see the Zone Object schema) so that the zone's aggregate `currentKw` can be reconciled against the underlying chargers by SIAM EV's internal systems. The EMS system only needs to store and return this array as opaque strings — it does not call the OCPP API, does not need to validate the IDs, and does not need connector-level detail (session state, connector status, etc.); that detail lives entirely in the OCPP charging system, which is out of scope for this document.
- **Out of scope for this handover.** OCPP charging telemetry, the Cargo Space Monitoring API, and the outbound webhook registry are separate SIAM EV–internal systems and are not included here. If a future feature requires the EMS system to call one of those APIs directly, SIAM EV will provide a supplementary interface specification at that time.

7 API Quick Reference

`Endpoint`	**Method**	**Description**
Facilities		
`/ems/facilities`	GET	List facilities
`/ems/facilities/{id}`	GET	Facility live snapshot
`/ems/facilities/{id}/readings`	GET	Time-series energy readings
`/ems/facilities/{id}/analytics`	GET	Demand KPIs for billing period
`/ems/facilities/{id}/demand-response`	POST	Activate / deactivate DR event
`/ems/facilities/{id}/power-limit`	PUT	Cap grid import power
Zones		
`/ems/facilities/{id}/zones`	GET	All zones: sensor + energy data (authoritative)
`/ems/facilities/{id}/zones/{zoneId}`	GET	Single zone detail
`/ems/facilities/{id}/zones/{zoneId}/readings`	GET	Zone sensor + energy time-series
`/ems/facilities/{id}/zones/{zoneId}/thresholds`	PUT	Update zone alert thresholds
WebSocket		
`/ems/ws`	WS	Live energy feed